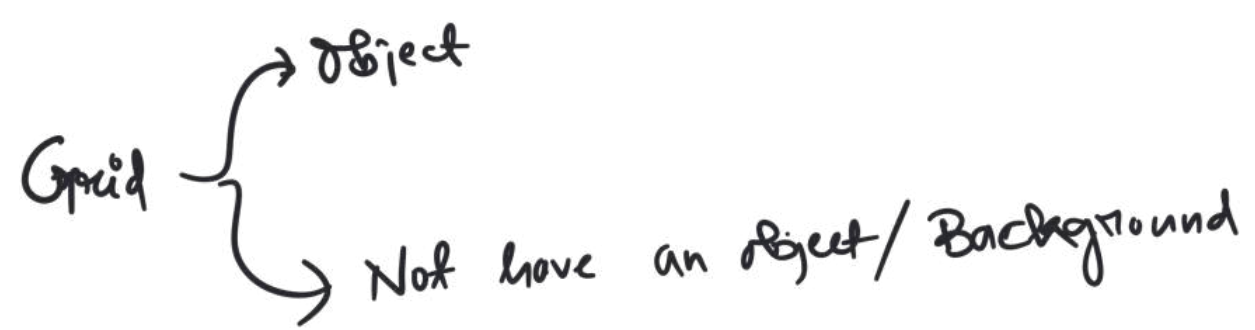


YOLO

Satya Pattnaik



{
Breadth
Depth } comfortable



$AB = \begin{bmatrix}$

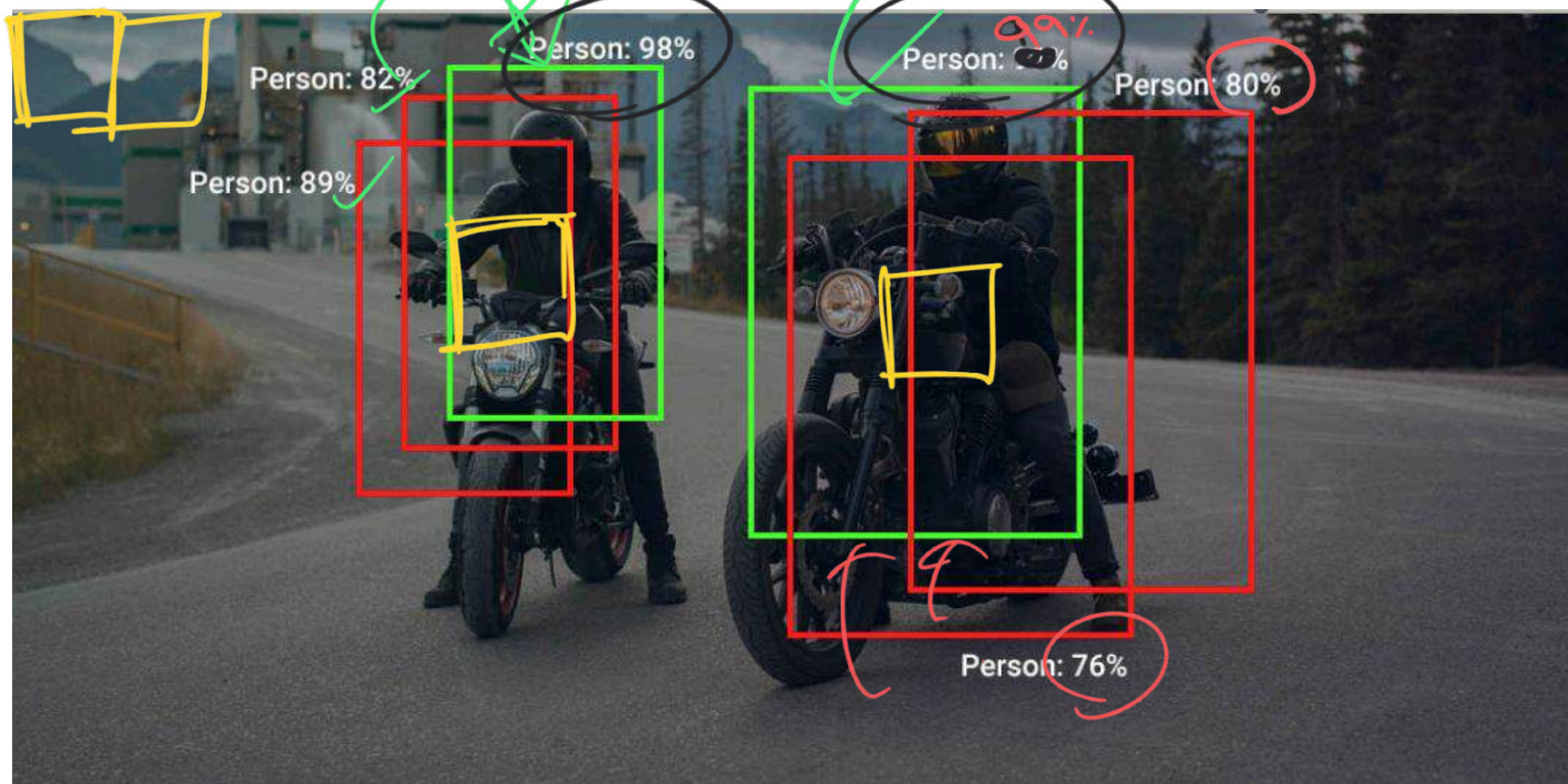
objectness score

$x_c \quad y_c \quad h \quad w$

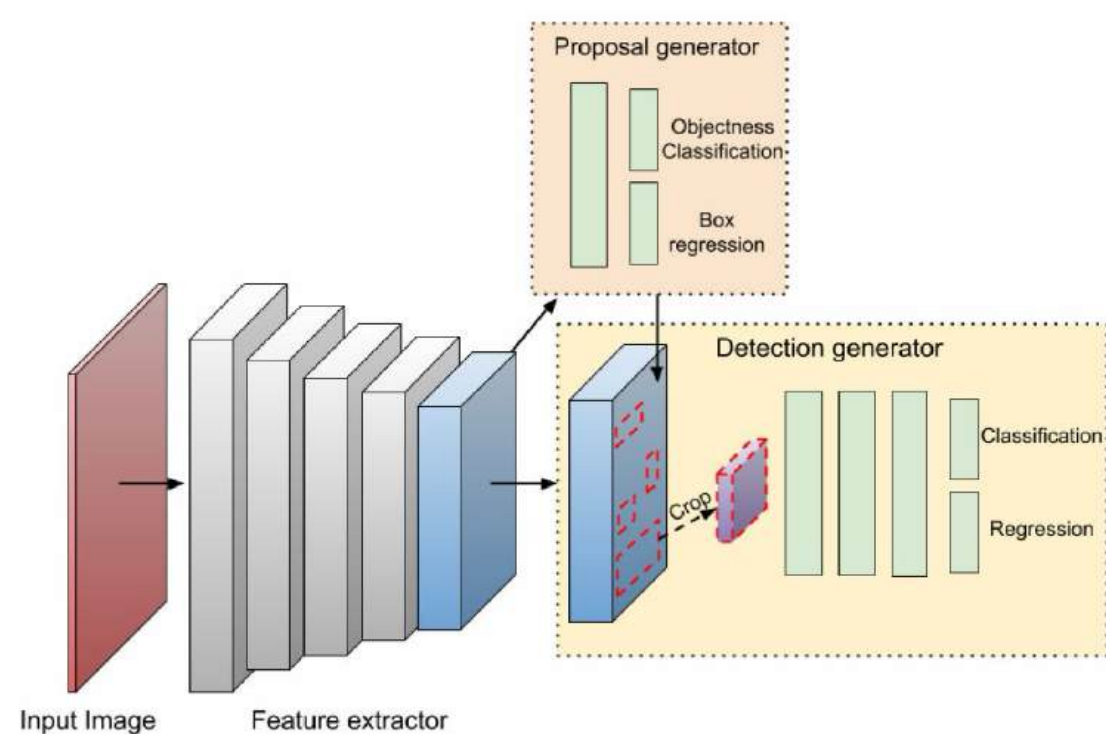


obj	x _c	y _c	h	w
99%	98%	81%	82%	80%
obj	obj			

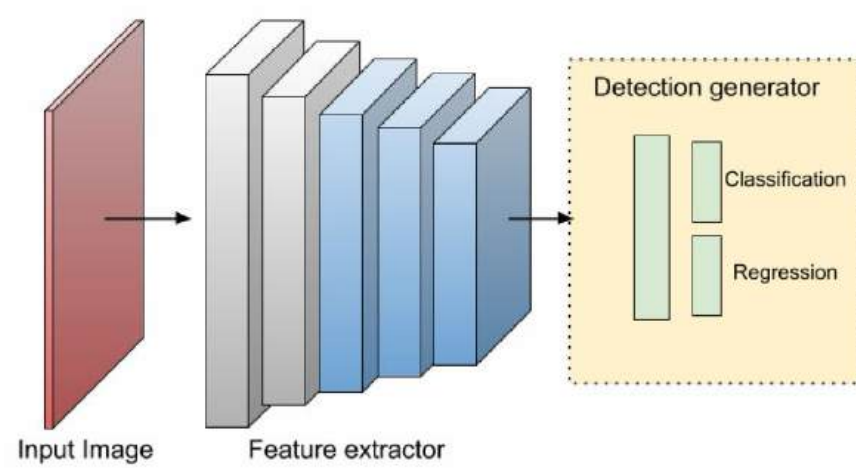
510000



YOLO



(a) Two-stage Faster R-CNN



(b) One-stage RetinaNet

~~Ref~~ → actual BB
 illustration $S \times S$ }
 7×7 Grid } not used actually



$S \times S$ grid on input

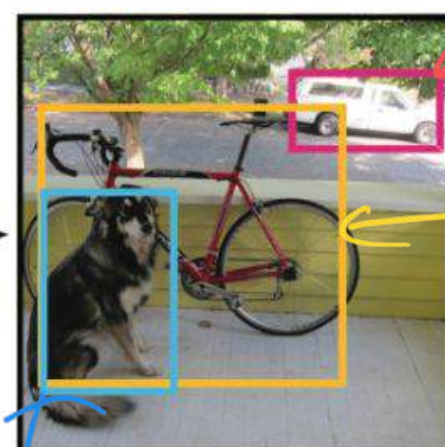


Bounding boxes + confidence



Class probability map

During train }
 { Bounding Box }
Centroid of the obj



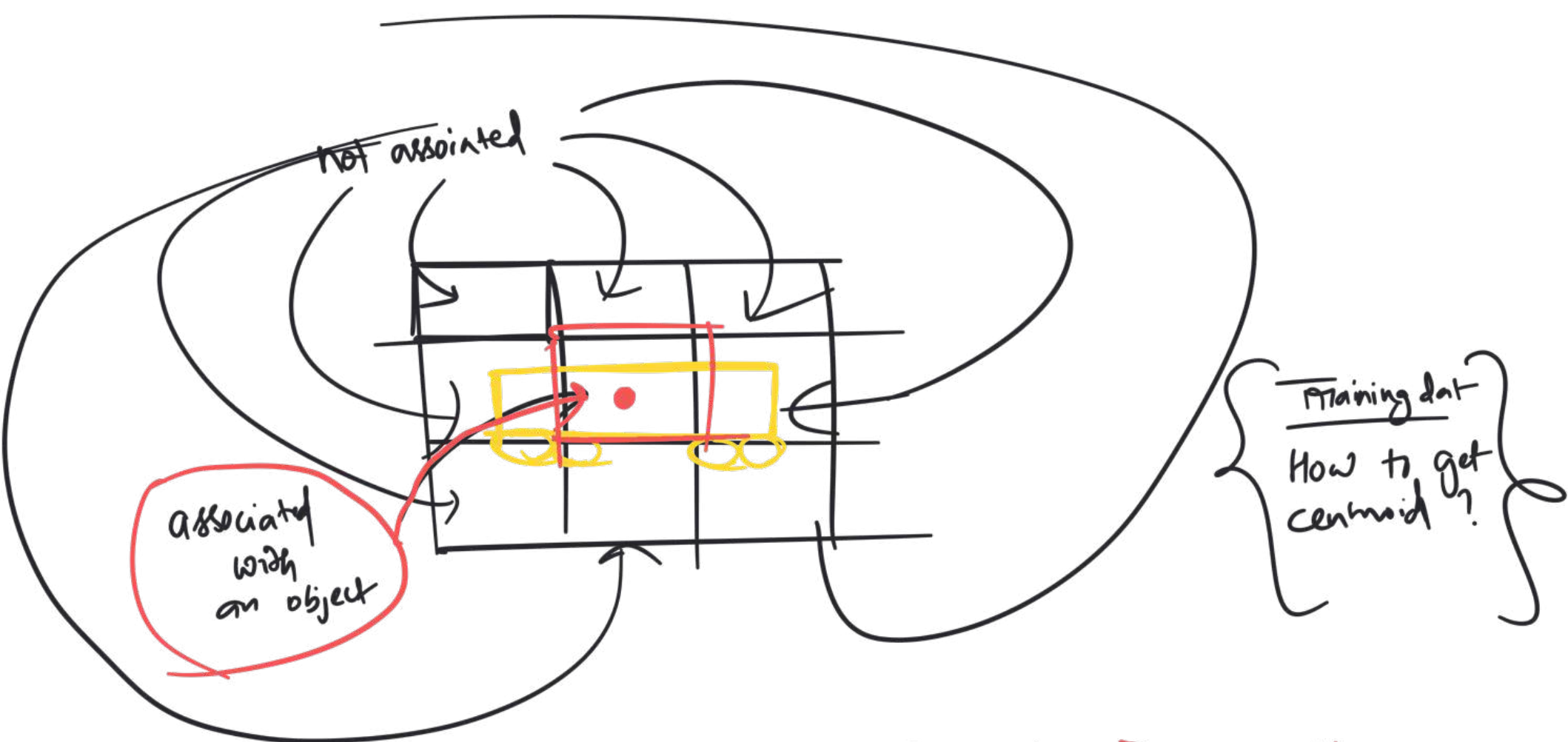
Final detections

Car

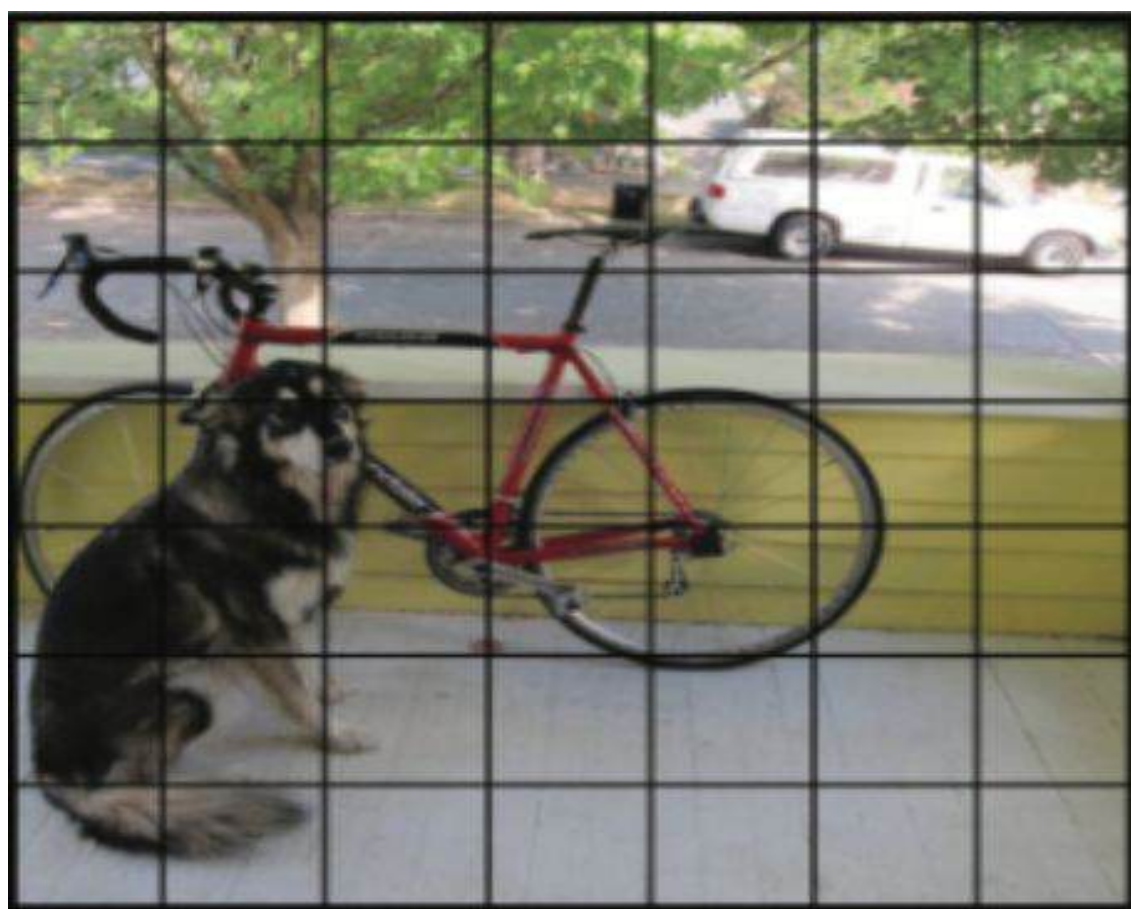
Cycle

Dog

How to associate a grid with an object?



Centroid — Prime minister



$S \times S$ grid on input

$$\text{Total grids} = 5^2$$

~~8x8~~ 5x5

1	2	

224/22413 Supervised

X

front of
image

each
grin

16x16s

2

•

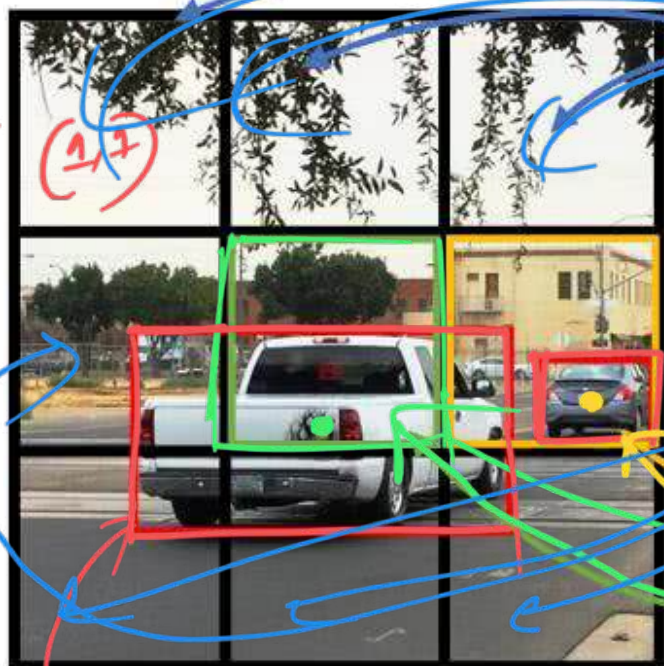
$\square_{S^2}$



→ $\begin{bmatrix} \text{obj} & x_c & y_c & h & w & \text{Core number} \\ \text{beg} & & & & & \square \square \square \end{bmatrix}$

8^2 such mappings ✓✓

$$\begin{bmatrix} \text{obj} & x_c & y_c & h & w & \text{dog} & \text{human} & \text{car} \\ 0 & ? & ? & ? & ? & 0 & 0 & 0 \\ & -1 & -1 & -1 & -1 & & & \end{bmatrix}$$
 } if grid doesn't have the obj



p_c	0	1	1
b_x	?	0.67	0.45
b_y	?	0.75	0.98
b_h	?	0.39	0.95
b_w	?	0.53	1.72
c_1	?	0	0
c_2	?	1	1
c_3	?	0	0

Is there an object?

Bounding box

Class labels

$$\begin{bmatrix} \text{obj} & x_c & y_c & h & w & \text{dog} & \text{human} & \text{car} \\ 1 & 0.45 & 0.98 & 0.95 & 1.72 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_c & y_c & h & w \\ 0.45 & 0.98 & 0.95 & 1.72 \end{bmatrix}$$
 } Training data
 human annotated

obj {

- dog
- human
- car

 } Training data
 no. of class in
 how rich your training data

YOLO in Easy Steps

Training

1] Divide img into $S \times S$ grid

2] $C \leftarrow$ Unique object classes
 $[obj/bbox \quad x_c \quad y_c \quad h \quad w]$ $\leftarrow$ "C" classes

$S \times S \times (5 + C)$

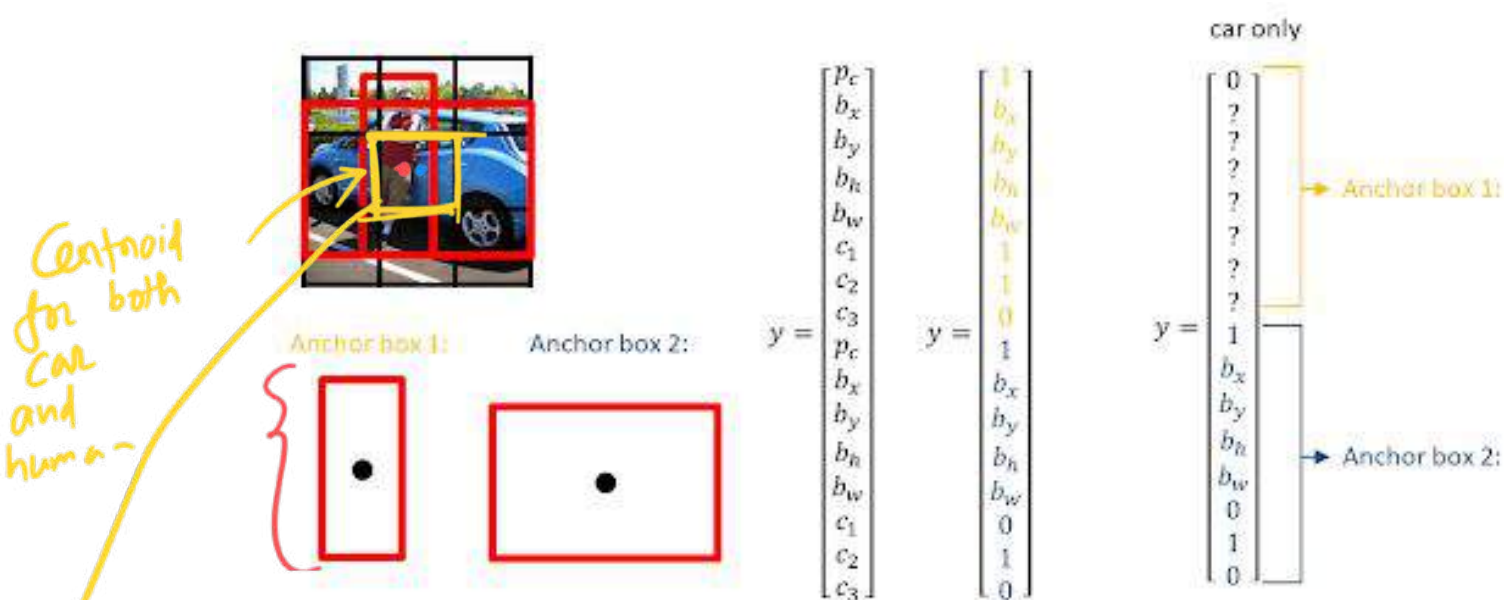
obj confidence score $\quad x_c \quad y_c \quad h \quad w$

$\left. \begin{matrix} dog \\ hum \\ car \end{matrix} \right\} C=3$

* What if we have multiple objects in the same grid?

Multiple Objects in Single Grid Cell

3 Anchor Boxes for grid
Predefined scales for grid



of anchor boxes = $\text{tp}(\text{hyper param})$
 # sizes of anchor boxes = Predefined
 Each grid = associated with multiple boxes

Objectness score

	Obj score	Car	Human	D	C	H
1	1	0	1	0	1	0

AB₁

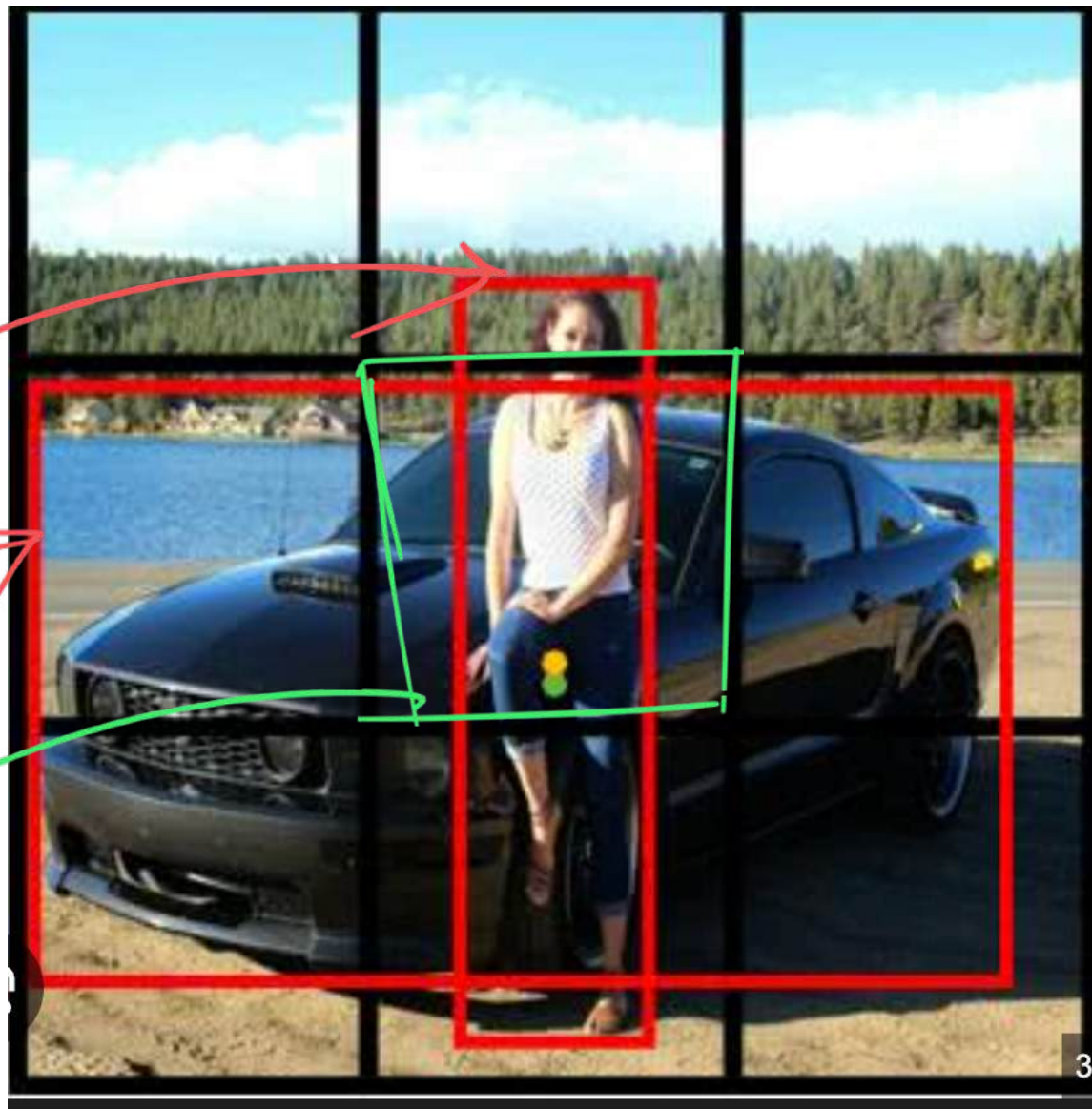
	Obj score	Human	D	C	H
1	1	0	0	0	1

AB₂

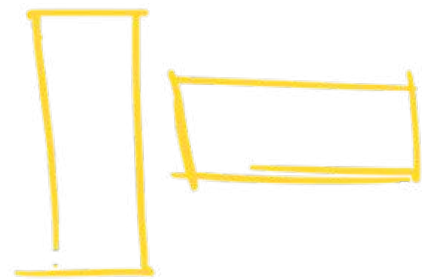
Anchor Boxes -
 for grid
 no. of AB's
Site of AB
 fixed
 predefined ✓
 not going to change
 not the same of Bounding Box

Bounding Boxes - Rect
 AB - Richards

Training data



Anchor Box
size $\neq$ Bounding Box



each grid

Brain

1/0

objectness score

Object present/not present

int

90%

x_c

y_c

h

w

1

2

...

C



class

How many total predictions in YOLO

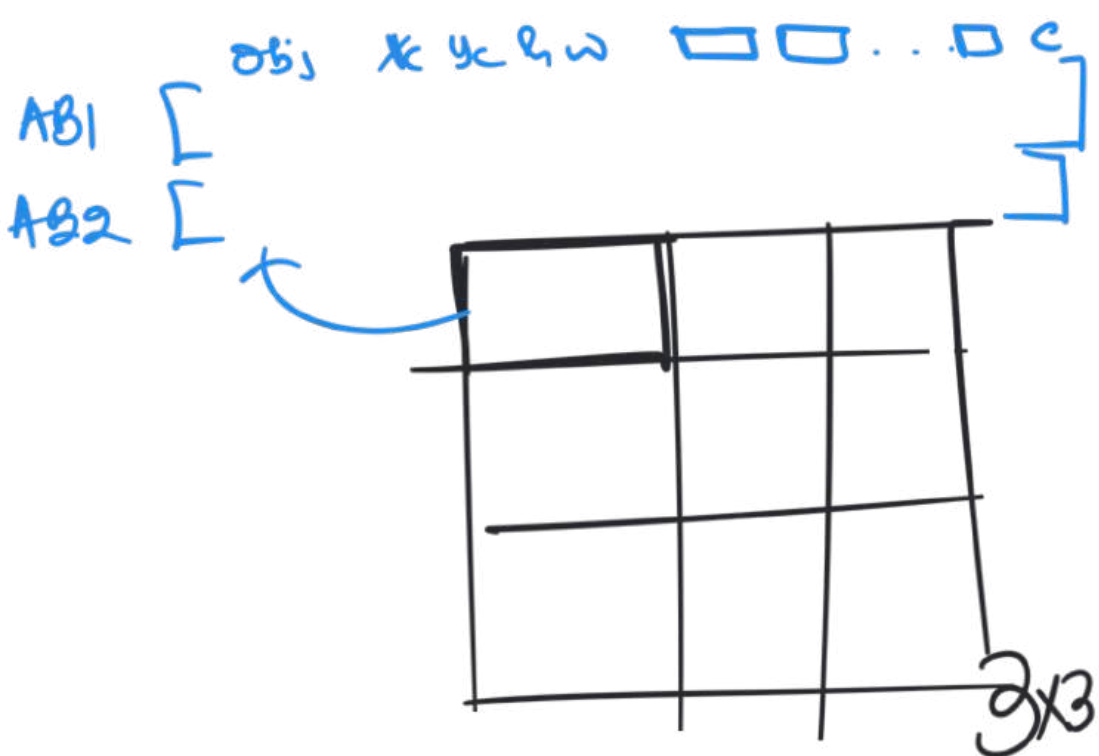
$S \times S$ grid

of ABs in each grid = 'AB'

total no. of object classes = C

no. of predictions for each image

$$S \times S \times \left[AB \times \left(\underset{\substack{\uparrow \\ \text{Objectness Score}}}{1} + \underset{\substack{\downarrow \\ \text{Bounding Box}}}{4} + \underset{\substack{\uparrow \\ \text{no. of classes}}}{C} \right) \right]$$



2 Anchor Boxes

$$3 \times 3 \times \left[a \times \left(\underset{\substack{\uparrow \\ \text{1 value} \\ \text{for obj} \\ \text{present} \\ \text{or not}}}{1} + \underset{\substack{\times \\ g \\ h \\ w}}{4} + c \right) \right]$$

10/10

you don't directly infer x_c y_c h w



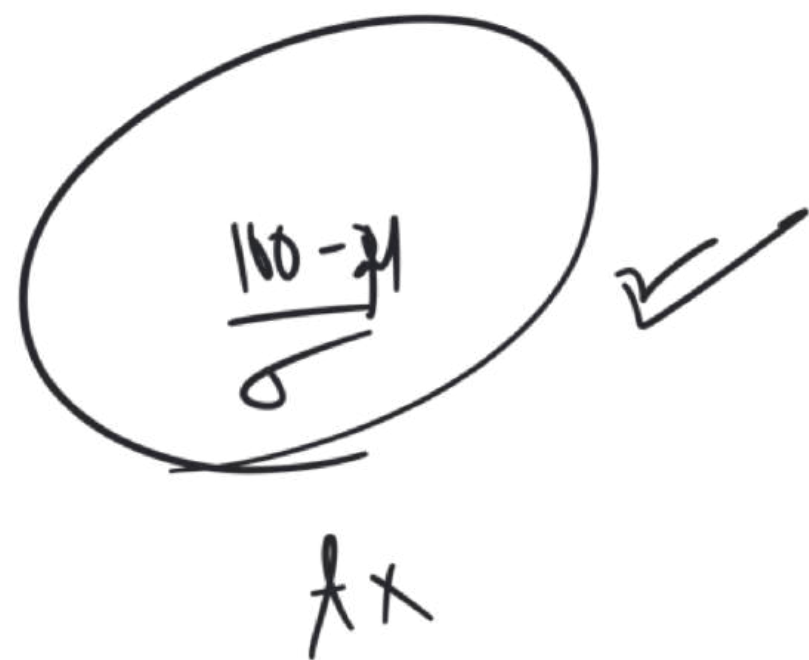
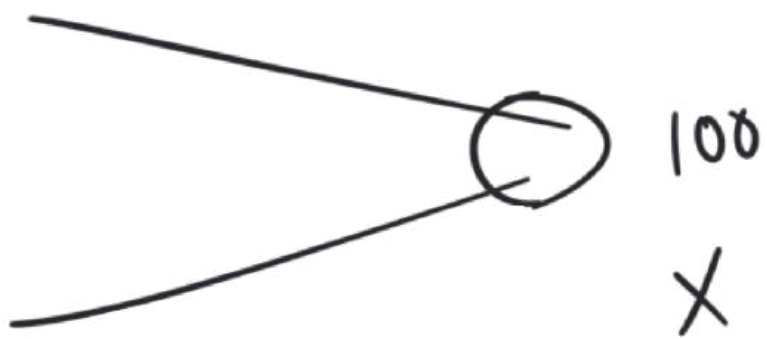
obj score



✓
transformation

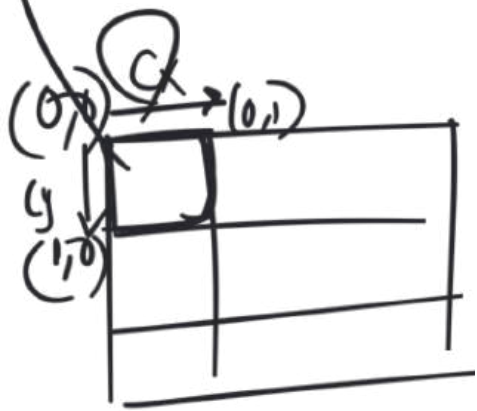
rx x_c y_c h w

Regression

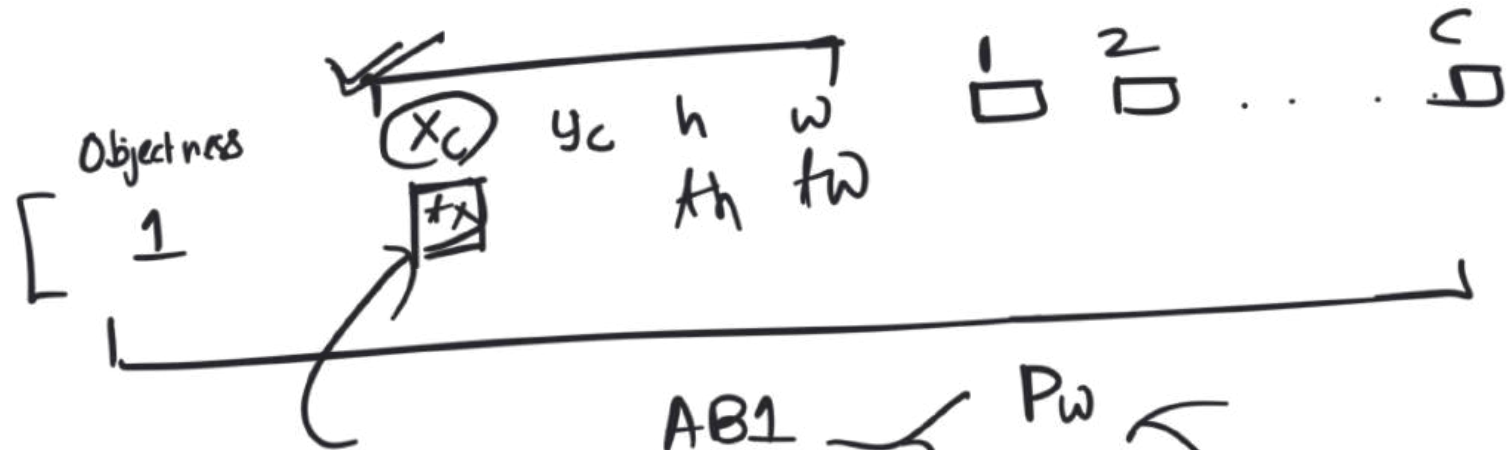


Training

grid =



3x3



AB1 { P_w
 P_h

$$t_x = \sigma^{-1}(X_c - C_x)$$

$$X_c = \sigma(t_x) + C_x$$

$$A_h = f(P_h)$$

$$t_w = f(\quad)$$

derive θ_x from x_c/b_x during train

derive x_c/b_x from θ_x during inference

$$\frac{x_c}{b_x}$$

Inference



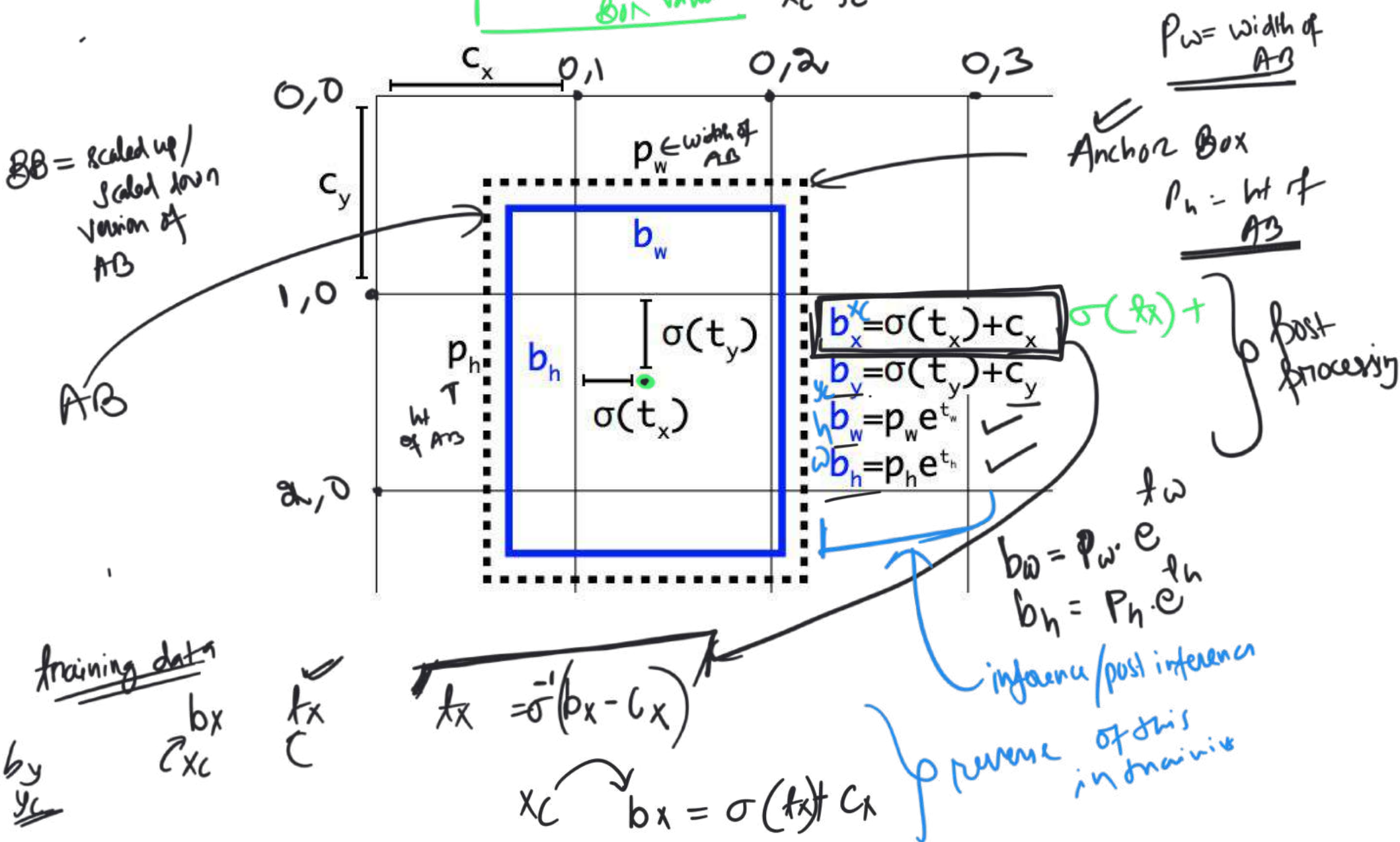
$$X_c = \sigma(t_x) + \underline{\underline{C_x}}$$



Each grid, for each Anchor Box $\sigma(b_i)$

Output from model $\checkmark t_x \checkmark t_y \checkmark t_w \checkmark t_h \leftarrow$ you predict

Actual Bounding Box value $\frac{b_x}{x_c} \frac{b_y}{y_c} \frac{b_w}{w} \frac{b_h}{w}$

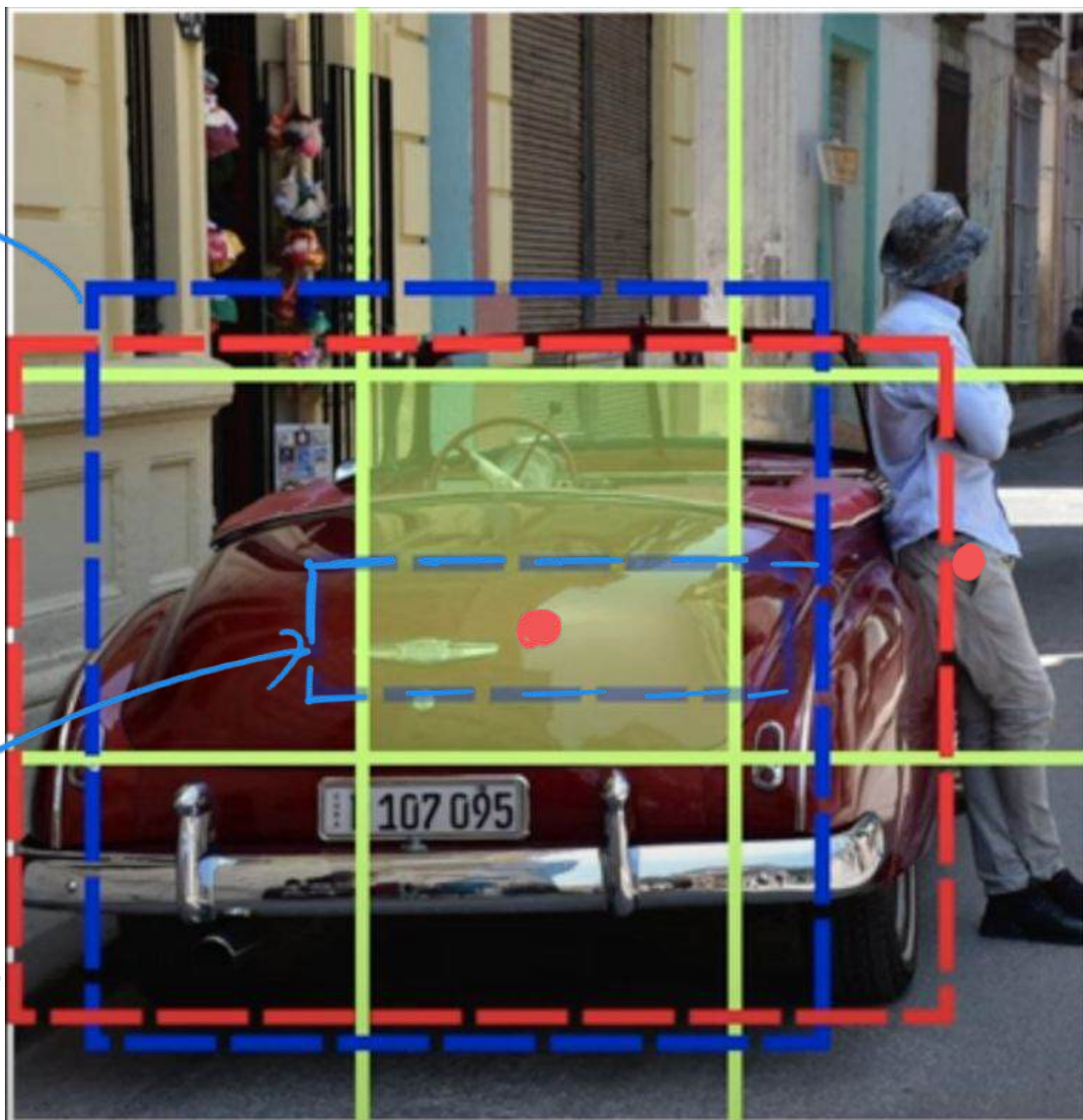


9:21

Anchor Box 1

Bounding Box of Car from data

Anchor Box 2



Intersection Over Union

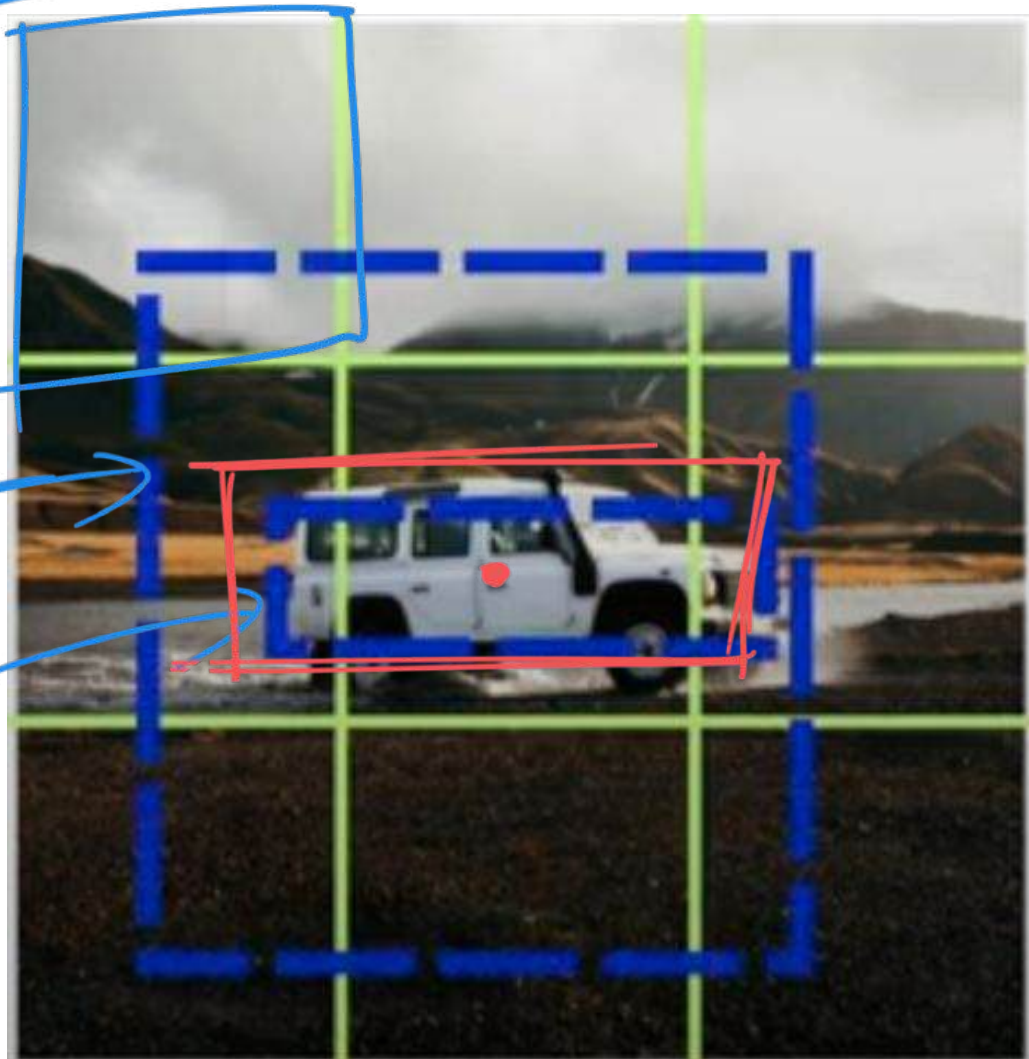
AB₁

AB₂

$$\left[\begin{array}{ccccccccc} 1 & x_c & y_c & h & w & D & C & H & \\ \text{Car} & & & & & 0 & 1 & 0 & \end{array} \quad \begin{array}{ccccccc} 0 & x_c & y_c & h & w & D & C & H & \\ & ? & ? & ? & ? & 0 & 0 & 0 & \end{array} \right]$$

gou fire breaker

high gou
AB2



9:45

	X	Y
1		ABI DHC AB2 $[0 \text{ } ? \text{ } ? \text{ } ? \text{ } ? \text{ } 0 \text{ } 0 \text{ } 0 \text{ } 0 \text{ } 7 \text{ } ? \text{ } ? \text{ } ? \text{ } ? \text{ } 0 \text{ } 0 \text{ } 8]$ $[1 \text{ } 0 \text{ } 5 \text{ } 0 \text{ } 5 \text{ } 0 \text{ } 5 \text{ } 0 \text{ } 5 \text{ } 0 \text{ } 0 \text{ } 1]$ <u>With highest IOU</u>
2		$[0 \text{ } 7 \text{ } 7 \text{ } . \text{ } 7 \text{ } .]$
5 ²		$[0 \text{ } 7 \text{ } 7 \text{ } . \text{ } 7 \text{ } .]$

$1_{ij}^{obj} \rightarrow 1$ if box j and cell i match,
0 otherwise

$1_i^{obj} \rightarrow 1$ if cell has object,
0 otherwise



$\lambda_{coord} = 5, 0$ } increasing
 $\lambda_{noobj} = 0, 5$

$$Loss_{yolo} = \lambda_{coord} \sum_{i=0}^{S^2} \sum_{j=0}^B 1_{ij}^{obj} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] + \lambda_{coord} \sum_{i=0}^{S^2} \sum_{j=0}^B 1_{ij}^{obj} \left[(\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] + \rightarrow \text{Bounding Box coord}$$

$$\sum_{i=0}^{S^2} \sum_{j=0}^B 1_{ij}^{obj} \left[(c_i - \hat{c}_i)^2 \right] + \lambda_{noobj} \sum_{i=0}^{S^2} \sum_{j=0}^B 1_{ij}^{noobj} \left[(c_i - \hat{c}_i)^2 \right] + \rightarrow \text{Confidence}$$

$$\sum_{i=0}^{S^2} 1_{ij}^{noobj} \sum_{C \in \text{classes}} \left[(p_i(C) - \hat{p}_i(C))^2 \right] \rightarrow \text{Classification}$$

$1_{ij}^{noobj} \rightarrow 1$ if box j and cell i no match,
0 otherwise

} Localization
low
} Obj conf
low
} Classification
low

λ_{coord}

$$\sum_{i=0}^{S^2} \sum_{j=0}^{AB} 1_{ij} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

{ actual x_c from ground truth }
 predicted x_c
 MSE

for each grid S^2 grids
 inside each grid, for each Anchor Box
 if the obj is present

2 Anchor Boxes - each grid 2 AB

$3 \times 3 = 9 \times 2$ AB

λ_{coord}

$$\sum_{i=0}^{S^2} \sum_{j=0}^{AB} 1_{ij} \left[(\sqrt{w_{ij}} - \sqrt{\hat{w}_{ij}})^2 + (\sqrt{h_{ij}} - \sqrt{\hat{h}_{ij}})^2 \right]$$

that grid i and Anchor Box j has the obj
 1 - if it has obj
 0 - if not

log 100

obj $\begin{cases} \text{present} \\ \text{absent} \end{cases}$

guid1 AB1  x_c, y_c, h, w

training data $\{ 1/0 \}$

probability of object present = 90%.

$$\sum_{i=0}^S \sum_{j=0}^{AB} 1_{ij} \left(c_i - \hat{c}_i \right)^2 \cdot (1 - 0.2)^2$$

prob from model

$(1 - 0.9)^2$

$(1 - 0.99)^2$
loss is minimized



$$\sum_{i=0}^2 \sum_{j=0}^{AB} 1_{ij}^{nobj} \left(C_i - \widehat{C}_i \right)^2$$

$\uparrow$ $\downarrow$
 0 0.5501

Softmax

$$\sum_{i=0}^S \sum_{c \in \text{Class}} 1_i^{\text{obj}} (p_i(c) - \hat{p}_i(c))^2$$

Q H C
1 0 0

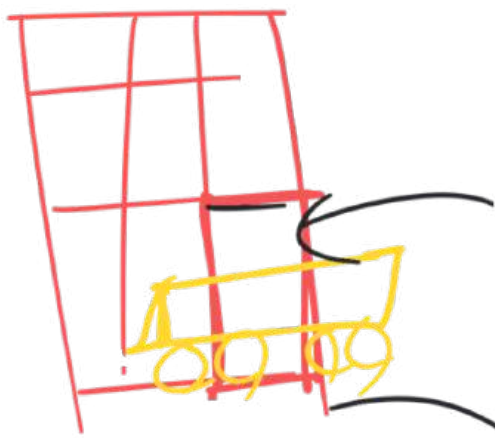


$$\underbrace{\left(\underset{\substack{\uparrow \\ \text{gt}}}{1} - \underset{\substack{\uparrow \\ p}}{0.6} \right)^2 + \left(\underset{\substack{\uparrow \\ \text{gt}}}{0} - \underset{\substack{\uparrow \\ p}}{0.2} \right)^2 + \left(\underset{\substack{\uparrow \\ \text{gt}}}{0} - \underset{\substack{\uparrow \\ p}}{0.2} \right)^2}$$

$$\begin{array}{c}
 5 \\
 \lambda_{obj} \sum_{\lambda=0}^5 \sum_{j=0}^{AB} 1_{ij} (C_i - \hat{C}_i)^2 + \lambda_{noobj} \sum_{i=0}^5 \sum_{j=0}^{AB} 1_{ij} (C_i - \hat{C}_i)^2 \\
 \underbrace{\hspace{10em}}_{5\%} \qquad \underbrace{\hspace{10em}}_{95\%}
 \end{array}$$

$$\begin{array}{l}
 \lambda_{obj} > \lambda_{noobj} \quad \boxed{} \\
 \lambda_{noobj} > \lambda_{obj} \quad \boxed{}
 \end{array}$$

nested for loop



$$\sum_{i=0}^2 \sum_{j=0}^2 A_{ij}$$

$$\sum_{i=0}^2 \sum_{j=0}^2 1_{ij}^{obj}$$

ground
truth
prob

D H C

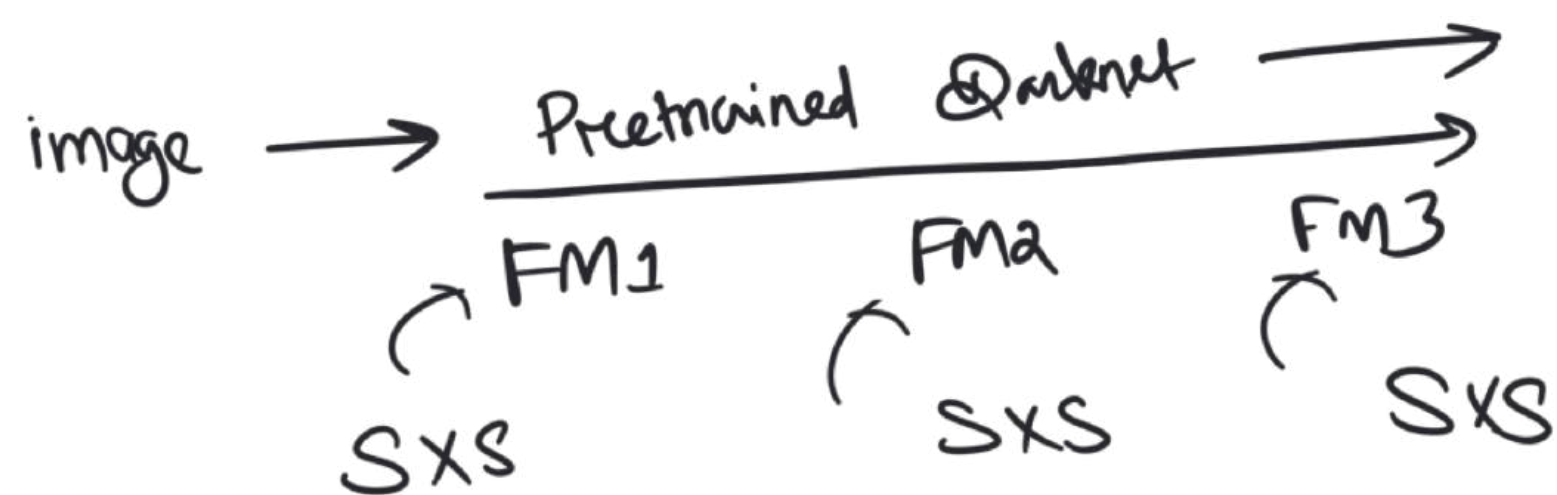
← true

0.2 0.2 0.6

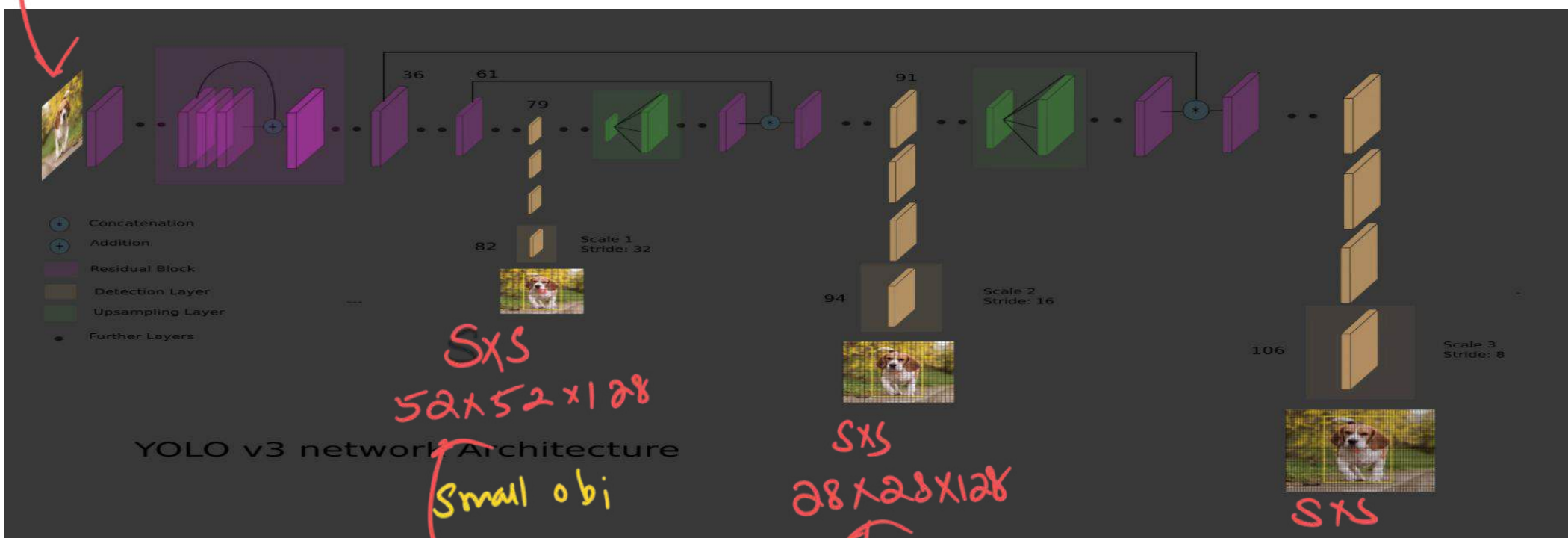
← model ✓

$$\left[(0 - 0.2)^2 + (0 - 0.2)^2 + (1 - 0.6)^2 \right]$$

~~Grid~~ SXS grids on original image ✓✓



SxS
 $224 \times 224 \times 3$



SxS
 $52 \times 52 \times 128$

Small obj

SxS
 $28 \times 28 \times 128$

Medium

SxS
 $8 \times 8 \times 128$ Large

good amount
 ✓
Dardenet
 =
ResNet
 V. G. Krizhevsky
AlexNet

	Type	Filters	Size	Output
	Convolutional	32	3×3	256×256
	Convolutional	64	$3 \times 3 / 2$	128×128
1x	Convolutional	32	1×1	
	Convolutional	64	3×3	
	Residual			128×128
	Convolutional	128	$3 \times 3 / 2$	64×64
2x	Convolutional	64	1×1	
	Convolutional	128	3×3	
	Residual			64×64
	Convolutional	256	$3 \times 3 / 2$	32×32
8x	Convolutional	128	1×1	
	Convolutional	256	3×3	
	Residual			32×32
	Convolutional	512	$3 \times 3 / 2$	16×16
8x	Convolutional	256	1×1	
	Convolutional	512	3×3	
	Residual			16×16
	Convolutional	1024	$3 \times 3 / 2$	8×8
4x	Convolutional	512	1×1	
	Convolutional	1024	3×3	
	Residual			8×8
	Avgpool		Global	
	Connected		1000	
	Softmax			

← FM₁
64x64

← FM₂
16x16

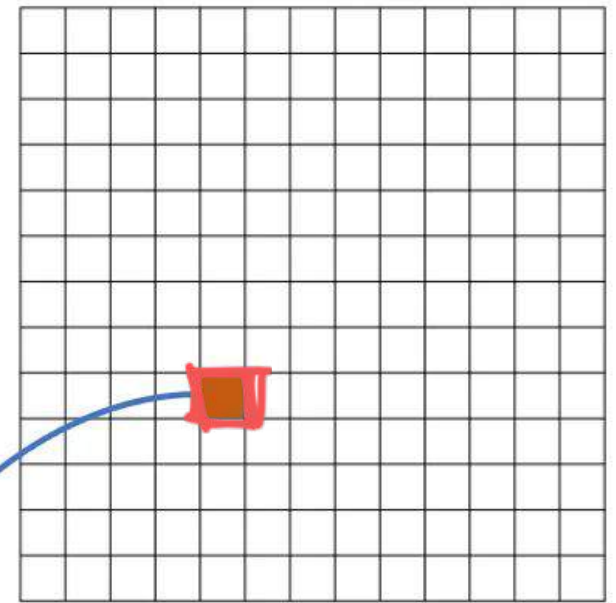
← FM₃
8x8

downsampling



Pre-processing Image
(416, 416, 3)

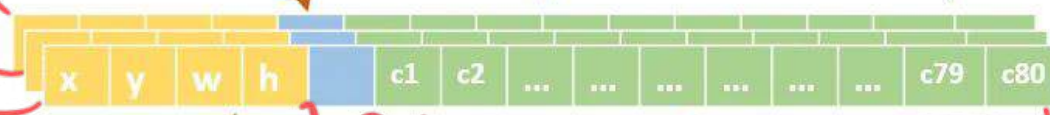
YOLO v3
Network



Encoding*
(13, 13, 3, (4+1+80))**

3 Anchor Boxes

Objectness
score



3 anchor boxes per
one grid cell***

Bounding box
offset: x, y, w, h

Class
probabilities

Objectness

$$13 \times 13 \times 3 \times (4 + 1 + 80)$$

no. of grids no. of AB xc yc hw obj score class cond prob

* In YOLO v3, the final output of detectors will be in shape 13x13, 26x26, 52x52. The example above is one of the output in 13x13 for illustration purposes only

** (13, 13, 3x(4+1+80)) are flatten into (13, 13, 255)

*** In YOLO v3, there are 3 anchor boxes per grid cell

